15/20

Al-Balqa Applied University
Faculty of Engineering Technology

General chemistry lab /50min
Midterm exam 20 mark

10/22/2022 – Sunday**

Student name [redacted]	Instructor: [illegible]	Section: 11 → 2 الثنين

[Helpful information: R = 0. 0821atm.L/mol. K; mwt: H_2O = 18 g/mol; 760 mmHg = 1 atm]

* Choose the correct answer for the following questions and transfer it to the table below:

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	A	C	C	A	A	B	A	D	D

Question	11	12	13	14	15	16	17	18	19	20
Answer	B	B	D	A	B	A	D	B	B	A

1- Which of the following statements is correct?

A. When dealing with alcohols heat with direct flame.
B. Return used chemicals to the stock bottles.
C. Hold your face directly over the container when you smell a chemical substance.
(D) It's not allowed to withdraw samples form the stock solution bottles with your own pipet.

2-The following set of data were obtained by student during measurement density of solid material, Calculate Mass of the solid: ----------:

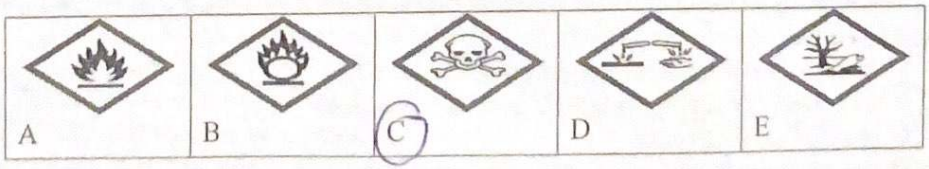
(A) 1.33 kg
B. 0.30 kg
C. 3.00 g
D. 13.3 g

$d = \frac{m}{v}$

$V = 66.5$

density of solid = 20 g/cm3
Initial volume of water in cylinder: 18.5 ml
Final volume of water in cylinder: 85.0 ml

3-The pictogram that correctly indicates a Toxic chemical hazard is:



$H_2O = 0,49$

anhydrous = 0,7 alum = 1,19

4- Given the following set of data:

- Mass of empty crucible = 12.1g.	- Mass of crucible + alum = 13.2 0g.
- Mass of crucible + anhydrous salt = 12.8g	- Molar mass of anhydrous salt = 258.0g/mol.

Calculate water of hydration (X) in the alum.

A- 13 B- 14 C- 8 D- 5

5- If (X= 6) and you have the following data calculate mass of hydrates salt

an = 1,02

A. 0.73 g
B. 1.75 g
C. 1.02 g
D. 0.68 g

Mass of empty crucible	5.21 g
Mass of crucible and anhydrous	6.23 g
Molar mass of anhydrous	150 g/mol
Molar mass of H_2O	18 g/mol

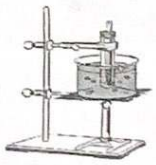
6- Calculate the temperature (in° C) of 1.4g of a vapor of a volatile Liquid (condensed vapor) (molar mass 90.0g/mol) which occupies a volume of 700.0 ml at a pressure of 720 mmHg:

A- 246 °C B- 225°C C- 519 C° D- 158 °C

7- The boiling points in lab experiment are indicate using a .....

A. Capillary tube
B. Test tube
C. Thermometer
D. Bunsen burner

8- In which experiment dose this apparatus used:



A-Boiling point identification
B-Density identification
C-Molecular weight of volatile liquid
D-Water of crystallization

9- The vapor of an unknown volatile liquid occupies flask at 100.0°C and 1 atm, if the molar mass of volatile liquid equal 50.0 g/mol. what is the density of this volatile liquid?

A- 6.1 g/L B- 1.2 g/L C-1.9 g/L D- 1.6 g/L

10- By heated to high temperature to drive off the water, this process called?

A. Boiling point
B. Dehydration
C. Hydrated
D. Crystallization

11- The boiling point of a liquid is dependent on pressure, which one of the following is correct

A. Yes, negative relationship with external pressure
B. Yes, Positive relationship with external pressure
C. No, does not effected
D. No, negative relationship with vapor pressure

12- Which of the following statement is incorrect?

A. $CH_3CH_2CH_2OH$ lower boiling point than $CH_3CH_2CH_2CH_2CH_2OH$

B. $CH_3CH_2CH_2OH$ lower boiling point than $CH_3CH_2CH_2CH_3$

C. CH_3OH higher boiling point than CH_3CH_3

D. $CH_3CH_2CH_2OH$ higher boiling point than $CH_3CH_2CH_2CH_3$

13- Which one of the following glass wear is more accurately?

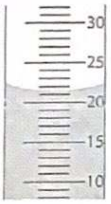
A. Beaker

B. Graduated Cylinder

C. Erlenmeyer Flask

D. Burette

14- What is the volume inside the cylinder?



A. 21.5 ml

B. 22.5 ml

C. 21.0 ml

D. 22.0 ml

15- One of the following salt is insoluble in H_2O (polar solvent)?

A. NaCl

B. $CaCO_3$

C. KI

D. K_2SO_4

16- In which solvent is the ionic salt more soluble?

A. Polar solvent

B. Non-polar solvent

C. Both polar and non-polar solvent

D. None of the above

17- A method used to vaporize a liquid into a fixed volume vessel at a measured temperature and Atmospheric pressure to measured molecular weight is called?

A. Synthesis method

B. Condensation method

C. Vaporization method

D. Dumas method

18-During molecular weight of volatile liquid experiment. In the end of experiment, we weight…….and the volume of vapor take by using…….?(Hint's; cover: Al foil and rubber).

A. mass of empty flask and cover - flask

B. mass of flask, cover and vapor - flask

C. mass of flask, cover and vapor - cylinder

D. mass of empty flask and cover - cylinder

19- During the hydrated salt Experiment how value of X effected if we put sample in desiccator?

A. Increase

B. Decrees

C. No effected

20- For most solid solutes, what happens to the solubility if the temperature decreases?

A. Solubility increases

B. Solubility decreases

C. Solubility stays the same



Good luck